24	**D**	Since ammeter reading is zero, there is also no current in the middle wire joining the circuits on the left and right. There is no potential difference between the two ends of this wire and there is no current exchange between the two circuits. 50 Ω and 100 Ω resistors are in series. *R* and 200 Ω resistors are in series. Potential difference across the 100 Ω and 200 Ω resistors is the same. $V_R = 24 - V_{200}$ $V_R = 24 - V_{100}$ $\frac{R}{R+200} \times 24 = 24 - \frac{100}{100+50} \times 12$ $\frac{R}{R+200} \times 24 = 16$ $R = 400\ \Omega$
25	**C**	The current in X produces a magnetic field along the circumference of coil Y in the